

CHANGJIN, Korea, Democratic People's Republic of

WMO: 470310

Lat: 40.367N

Lon: 127.250E

Elev: 1080

StdP: 89.01

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.6	-24.5	-29.3	0.3	-25.7	-27.3	0.4	-23.6	9.5	-16.0	8.9	-15.4	1.0	0	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	26.0	19.7	24.6	19.1	23.2	18.4	21.4	24.3	20.5	23.2	19.7	22.1	2.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.6	17.4	23.3	19.6	16.4	22.1	18.8	15.6	21.2	68.1	24.4	64.7	23.3	61.5	22.3	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.8	7.6	6.4	-30.6	28.5	2.5	1.4	-32.4	29.6	-33.8	30.4	-35.2	31.2	-37.0	32.2
			-30.2	22.7	2.6	1.0	-32.1	23.4	-33.6	24.0	-35.0	24.5	-36.9	25.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-14.3	-11.5	-4.4	3.4	10.3	15.0	18.6	18.3	12.6	5.8	-2.9	-11.6
	DBStd	12.14	4.48	4.78	5.17	4.07	3.39	2.45	2.54	2.55	2.89	3.44	5.03	4.86
	HDD10.0	3239	754	602	446	199	38	1	0	0	9	135	386	669
	HDD18.3	5538	1013	835	704	448	251	102	28	32	172	388	636	927
	CDD10.0	814	0	0	0	2	46	151	266	257	88	5	0	0
	CDD18.3	70	0	0	0	0	1	2	36	31	1	0	0	0
	CDH23.3	283	0	0	0	0	17	35	137	92	2	0	0	0
	CDH26.7	25	0	0	0	0	2	1	14	8	0	0	0	0
Wind	WSAvg	1.7	1.8	1.7	1.9	2.1	2.1	1.5	1.2	1.2	1.2	1.6	1.8	1.8
Precipitation	PrecAvg	783	7	14	16	41	66	97	209	159	96	38	31	14
	PrecMax	1164	25	49	58	90	147	234	539	477	254	116	106	62
	PrecMin	450	1	1	4	3	7	29	33	45	11	2	2	1
	PrecStd	199	5	15	11	22	31	50	109	101	69	24	24	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	4.3	12.2	20.1	25.3	26.5	28.5	27.7	23.1	18.8	11.3	2.4
		MCWB	-0.9	1.3	5.7	10.3	14.3	16.7	21.5	21.3	16.4	11.5	6.4	0.4
	2%	DB	-2.3	1.2	8.8	16.4	22.4	24.2	26.2	25.5	21.0	15.9	8.3	0.3
		MCWB	-4.2	-1.0	3.8	8.9	12.7	16.0	20.7	20.4	15.1	10.1	4.6	-1.1
	5%	DB	-4.4	-0.9	5.8	13.9	20.3	22.4	24.8	24.1	19.7	13.9	6.1	-1.8
		MCWB	-5.9	-3.1	1.9	7.5	12.1	15.9	20.2	19.9	14.7	8.8	3.7	-3.1
	10%	DB	-6.2	-2.8	3.4	11.4	18.0	20.8	23.4	22.7	18.1	11.8	4.1	-3.7
		MCWB	-7.4	-4.6	0.3	6.1	11.2	15.7	19.6	19.1	13.9	8.0	2.1	-5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.6	1.8	6.7	12.9	16.3	19.6	22.9	22.6	18.9	13.1	7.7	1.0
		MCDB	0.4	3.7	10.5	17.9	21.9	23.3	26.3	26.1	21.5	16.7	10.2	2.1
	2%	WB	-4.0	-0.7	4.1	9.8	14.5	18.1	21.8	21.3	16.9	11.2	5.4	-0.8
		MCDB	-2.6	1.3	7.7	14.9	20.3	21.6	24.6	24.2	19.2	14.2	7.2	0.2
	5%	WB	-5.7	-2.9	2.1	8.0	13.2	17.1	20.9	20.5	15.6	9.7	3.9	-3.1
		MCDB	-4.5	-1.1	5.5	13.2	18.4	20.6	23.6	23.1	18.0	12.7	5.7	-1.9
	10%	WB	-7.4	-4.6	0.5	6.4	11.8	16.2	20.0	19.7	14.6	8.6	2.3	-4.9
		MCDB	-6.2	-2.9	3.1	10.9	16.5	19.4	22.6	22.0	17.2	11.3	3.9	-3.8
Mean Daily Temperature Range	5% DB	MDBR	12.9	13.3	11.5	11.8	12.8	10.5	7.9	8.1	10.5	11.6	9.7	11.3
		MCDBR	13.4	14.2	13.5	16.9	17.8	14.3	10.4	9.8	12.1	14.8	12.0	11.5
		MCWBR	11.7	11.8	8.8	10.4	9.1	7.7	5.6	5.4	6.7	9.0	8.6	10.0
	5% WB	MCDBR	13.0	13.4	12.5	15.5	14.9	11.0	8.8	8.4	9.9	12.8	10.3	11.0
		MCWBR	11.6	11.4	8.7	10.3	9.0	7.0	5.4	5.3	6.5	8.7	8.0	9.8
Clear-Sky Solar Irradiance	taub	0.323	0.390	0.461	0.529	0.553	0.595	0.538	0.479	0.424	0.428	0.374	0.326	
	taud	2.079	1.923	1.813	1.748	1.750	1.717	1.927	2.085	2.162	2.024	2.078	2.115	
	Ebn at Noon	805	791	775	753	747	715	754	790	809	744	735	767	
	Edh at Noon	112	154	192	220	224	232	187	155	134	138	113	100	
All-Sky Solar Radiation	RadAvg	2.59	3.51	4.43	5.02	5.49	5.02	4.40	4.41	4.24	3.43	2.38	2.17	
	RadStd	0.13	0.21	0.30	0.38	0.50	0.48	0.55	0.45	0.41	0.25	0.29	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page